

Lecture 2: Policy Iteration

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Markov Property

$$\mathbb{P}[S_{t+1}|S_t, A_t] = \mathbb{P}[S_{t+1}|S_0, A_0, S_1, A_1 \dots, S_t, A_t]$$

$$\mathbb{P}[R_t|S_t, A_t] = \mathbb{P}[R_t|S_0, A_0, S_1, A_1 \dots, S_t, A_t] = 1$$

Markov Decision Process

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Markov Decision Process $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{P}_0, \mathcal{R}, \gamma \rangle$

- ▶ $\mathcal{S}$ is a **finite** ($|\mathcal{S}| = n$) state space
- ▶ $\mathcal{A}$ is a **finite** ($|\mathcal{A}| = m$) action space
- ▶ $\mathcal{P}$ is a **known** transition probability function

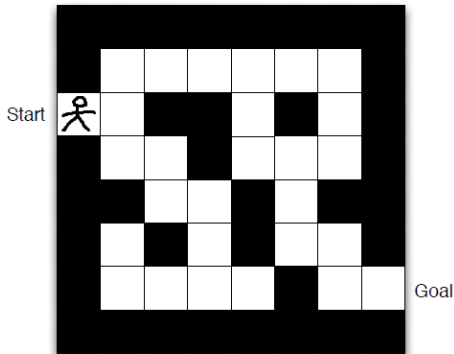
$$\mathcal{P}(s'|s, a) = \mathbb{P}[S_{t+1} = s'|S_t = s, A_t = a]$$

- ▶ $\mathcal{P}_0$ is a **known** initial state probability function
- ▶ $\mathcal{R}$ is a **known** reward function

$$\mathcal{R}(s, a) = R_t \quad \Leftrightarrow \quad \mathbb{P}[R_t|S_t = s, A_t = a] = 1$$

- ▶ $\gamma \in [0, 1]$ is a discount coefficient

Example: Maze



- States: white cells
- Actions: $\uparrow$, $\rightarrow$, $\downarrow$, $\leftarrow$
- Rewards: -1 for each step, 0 if Goal
- Final state: Goal cell

Main RL Algorithm Steps

- **Policy Evaluation**

Estimate v_π or q_π

- **Policy Improvement**

Update the policy: $\pi_{\text{new}} > \pi_{\text{old}}$

How to solve Bellman Expectation Equation?

Bellman Expectation Equation for v_π

$$v_\pi(s) = \sum_a \pi(a|s) \left(\mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) v_\pi(s') \right)$$

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$$v_\pi = \begin{pmatrix} v_\pi(s_1) \\ \vdots \\ v_\pi(s_n) \end{pmatrix}, \mathcal{R}_\pi = \begin{pmatrix} \mathcal{R}_\pi(s_1) \\ \vdots \\ \mathcal{R}_\pi(s_n) \end{pmatrix}, \mathcal{P}_\pi = \begin{pmatrix} \mathcal{P}_\pi(s_1, s_1) & \dots & \mathcal{P}_\pi(s_1, s_n) \\ \vdots & \ddots & \vdots \\ \mathcal{P}_\pi(s_n, s_1) & \dots & \mathcal{P}_\pi(s_n, s_n) \end{pmatrix}$$

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Theorem

If $\gamma < 1$ then there exists a unique solution v_π of Bellman Expectation Equation.

Iterative Policy Evaluation (Fixed-Point Iteration)

Let $\pi; v^0(s), s \in \mathcal{S}, K \in \mathbb{N}$.

For each $k \in \overline{0, K}$, do

$$v^{k+1}(s) = \sum_a \pi(a|s) \left(\mathcal{R}(s, a) + \gamma \sum_{s'} \mathcal{P}(s'|s, a) v^k(s') \right), \quad s \in \mathcal{S}$$

or

$$v^{k+1} = \mathcal{R}_\pi + \gamma \mathcal{P}_\pi v^k$$

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$$v^{k+1} = \mathcal{R}_\pi + \gamma \mathcal{P}_\pi v^k$$

Theorem

$v^k \rightarrow v_\pi$, $k \rightarrow \infty$. Convergence rate $O(mn^2)$

Partially Order for Policies

$$\pi' \geq \pi \quad \Leftrightarrow \quad v_{\pi'}(s) \geq v_{\pi}(s), \quad \forall s \in \mathcal{S}$$

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Greedy Policy Improvement

$$\pi'(a|s) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} q_{\pi}(s, a') \\ 0, & \text{otherwise} \end{cases}$$

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Policy Improvement Theorem

Let π . If π' is defined by Greedy Policy Improvement then

$$\pi' \geq \pi$$

Policy Iteration Algorithm

Let π^0 and $L, K \in \mathbb{N}$.

For each $k \in \overline{0, K}$, do

- **Policy evaluation.** Iterative Policy Evaluation

$$v^{l+1} = \mathcal{R}_{\pi^k} + \mathcal{P}_{\pi^k} v^l, \quad l \in \overline{0, L-1}.$$

Define $q^L(s, a)$ by $v^L(s)$

- **Policy improvement.** Greedy Policy Improvement

$$\pi^{k+1}(a|s) = \begin{cases} 1, & \text{if } a \in \operatorname{argmax}_{a' \in \mathcal{A}} q^L(s, a') \\ 0, & \text{otherwise} \end{cases}$$

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Theorem

$\pi^k \rightarrow \pi_*$, $k \rightarrow \infty$. Convergence rate $O(mn^2)$

QUESTIONS?